E-commerce Profitability Analysis

A business-focused analytics case study using Excel and Power BI

Total Sales	Total Profit	Profit Margin
2,326,534.35	292,296.81	12.56%
Distinct Orders	**Customers**	**Order-line rows**
5,111	804	10,194

Business question

Where is the company making and losing money? The project examines product profitability, discount patterns, regional performance, customer segments, shipping modes, and yearly trends.

Tools: Microsoft Excel 2019 and Microsoft Power BI

1. Executive Summary

This project analyzes 10,194 order-line records to understand sales and profitability across products, discount levels, regions, customer segments, shipping modes, and years. Excel was used for profiling and exploratory analysis; Power BI was used to present KPI cards, charts, slicers, and detailed matrices.

Headline results: sales were 2,326,534.35, profit was 292,296.81, and overall profit margin was 12.56%. Technology generated the highest category profit. Furniture had a 2.61% margin. Tables, Bookcases, and Supplies were the three loss-making sub-categories. West had the highest regional profit and margin; Central had the lowest regional margin. Discount levels of 30% and above were associated with negative total profit in this dataset.

Interpretation caution: these are descriptive associations, not proof that discounting alone caused losses. Product mix, volume, region, and other factors may influence profitability.

2. Business Objectives

• Measure sales, profit, and profit margin.
• Identify profitable and loss-making categories/sub-categories.
• Explore discount-level profitability and investigate selected loss patterns.
• Compare regions, customer segments, shipping modes, and years.
• Communicate findings in an interactive Power BI report.

3. Dataset and Grain

The working Orders table contains 10,194 rows and 21 original fields, plus a Postal Code Corrected helper field. Each row represents an order line/product line, not a unique order. The People table contains 4 rows; Returns contains 296 rows and 296 unique Order IDs. The data covers the United States and Canada, with 4 regions and 59 states/provinces. There are 5,111 distinct Order IDs, 804 distinct Customer IDs, and 1,862 Product IDs.

Order dates range from 3 Jan 2023 to 30 Dec 2026; ship dates range from 7 Jan 2023 to 5 Jan 2027. The exact source filename/version and download date were not recorded and should be added if recoverable.

4. Data Quality and Preparation

Recorded checks found no missing values in original Orders fields, no exact duplicate order lines, no zero/negative sales, no discount values outside 0–80%, no zero quantities, and no ship dates before order dates. TRIM-based checks across six categorical fields found no detected leading/trailing spaces or obvious label variations. Profit values below or equal to zero were retained because they are important to profitability analysis.

A helper field named Postal Code Corrected adds a leading zero to four-character US postal codes while preserving the original postal-code field and retaining Canadian alphanumeric codes. This standardizes formatting; it does not independently validate postal codes. Raw data was kept separate from the working workbook.

5. KPI Definitions

• Total Sales = sum of Sales.
• Total Profit = sum of Profit.
• Profit Margin = Total Profit ÷ Total Sales (12.56% overall).
• Total Orders = distinct Order ID count (5,111), not row count.
• Total Customers = distinct Customer ID count (804).
• Loss-making lines = rows with Profit < 0 (1,901).
• Zero-profit lines = rows with Profit = 0 (65).

6. Findings

6.1 Category performance

Category	Sales	Profit	Margin
Furniture	754,747.76	19,730.00	2.61%
Office Supplies	731,893.31	126,023.44	17.22%
Technology	839,893.28	146,543.38	17.45%
Overall	2,326,534.35	292,296.81	12.56%

Technology had the highest category sales and profit. Furniture's 2.61% margin was substantially below the overall margin, so it is a priority for further investigation.

6.2 Product sub-categories

Tables recorded –17,753.21 profit, Bookcases –3,632.07, and Supplies –1,171.39. Together they had 370,106.88 in sales and –22,556.67 profit, for an approximate combined margin of –6.10%. Copiers had the highest listed sub-category profit at 56,093.94; Paper had a 43.39% margin.

6.3 Regional performance

Region	Sales	Profit	Margin
West	739,813.61	110,798.82	14.98%
East	691,828.17	94,883.26	13.71%
South	391,721.91	46,749.43	11.93%
Central	503,170.67	39,865.31	7.92%

West led on sales, profit, and margin. Central had the lowest regional margin. Central Furniture was the only loss-making region-category combination identified, with –2,802.21 profit and –1.70% margin.

6.4 Customer segments

Consumer: sales 1,170,659.79, profit 136,371.45, margin 11.65%. Corporate: sales 715,806.13, profit 94,249.64, margin 13.17%. Home Office: sales 440,068.43, profit 61,675.73, margin 14.02%. Consumer generated the most sales and profit; Home Office had the highest margin.

6.5 Discount-level profitability

The 0% discount level recorded 326,718.59 profit and a 29.56% margin. Discount levels of 30%, 32%, 40%, 45%, 50%, 60%, 70%, and 80% each recorded negative total profit. The 70% discount level had the largest total loss (–40,300.57); the 80% level had the lowest margin (–180.01%). These aggregate patterns are observational and do not establish causality.

6.6 Root-cause investigation

A drill-down focused on Central Furniture and Furnishings at a 60% discount. Central Furnishings had 207 rows, sales of 15,284.36, profit of –3,915.52, and a –25.62% margin. Within this, 139 rows at 60% discount recorded 6,658.70 sales and –5,950.96 profit (–89.37% margin). West Furnishings had a 25.55% overall margin and no 60% discount records in the recorded comparison.

Product-level inspection found several loss-making chair mats in the Central Furnishings/60% discount group. The five worst listed products contributed about 2,203.46 in losses, roughly 37% of that group's net loss. This is an investigative lead, not proof of cause; more product-cost, pricing, and operational context would be needed.

6.7 Yearly performance

Year	Sales	Profit	Margin	YoY Sales	YoY Profit
2023	494,040.21	51,684.30	10.46%	—	—
2024	472,993.03	62,020.97	13.11%	–4.26%	+20.00%
2025	613,933.58	82,665.20	13.46%	+29.80%	+33.29%
2026	745,567.53	95,926.35	12.87%	+21.44%	+16.00%

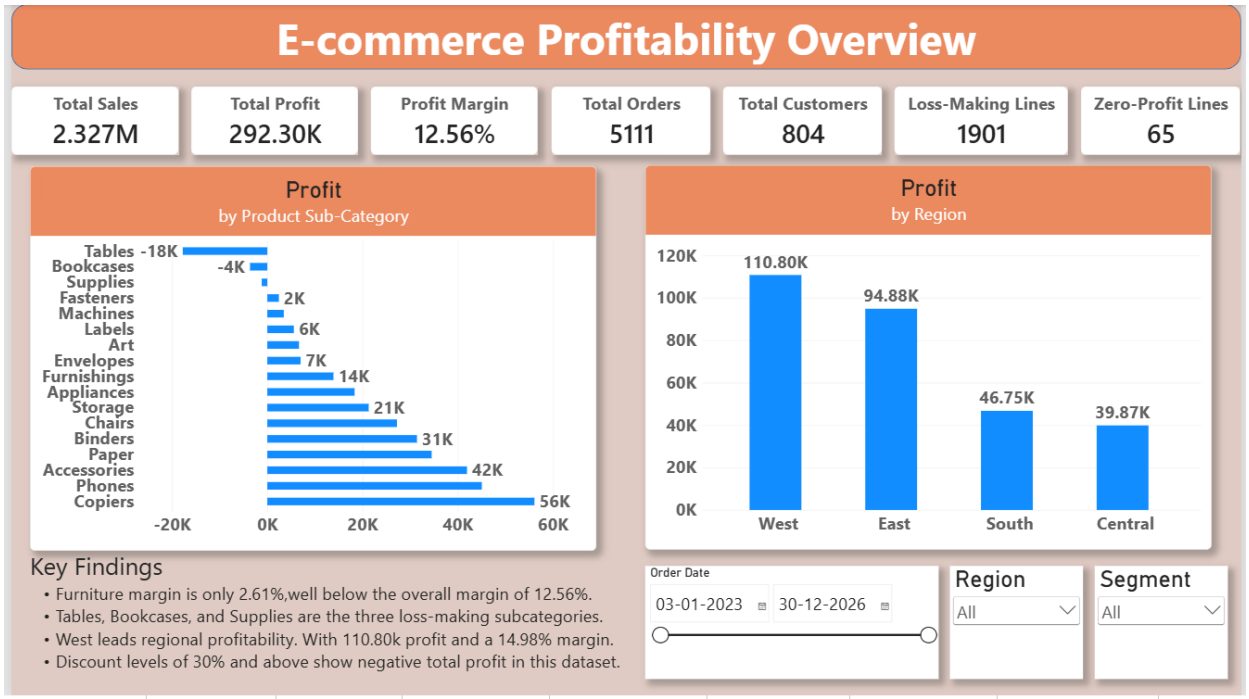
Sales and profit reached their highest recorded values in 2026. Margin peaked in 2025 at 13.46%, then declined to 12.87% in 2026. The recorded date range ends on 30 December 2026, so the bounds alone do not prove full-year completeness.

6.8 Shipping mode

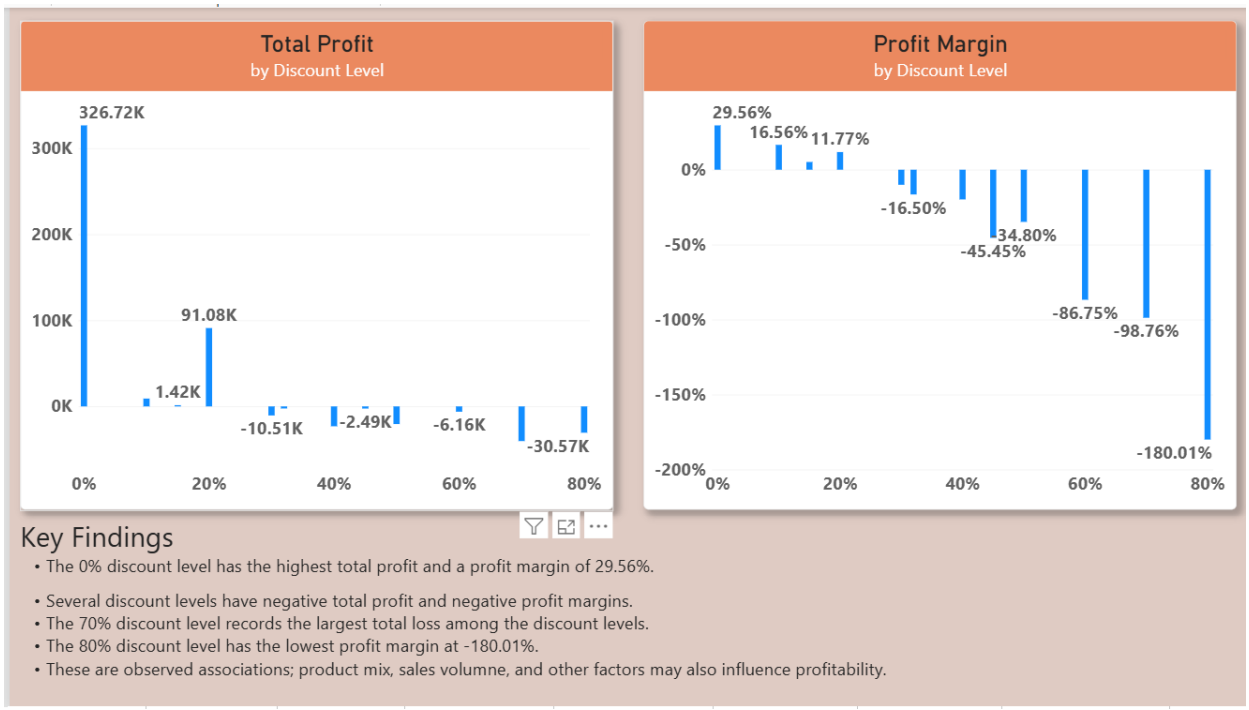
First Class had the highest margin at 13.93%. Standard Class generated the most sales (1,378,840.55) and profit (168,161.21), with a 12.20% margin. The recorded analysis does not document a separate shipping-cost field, so these results should not be interpreted as delivery-cost efficiency.

7. Power BI Report and Screenshots

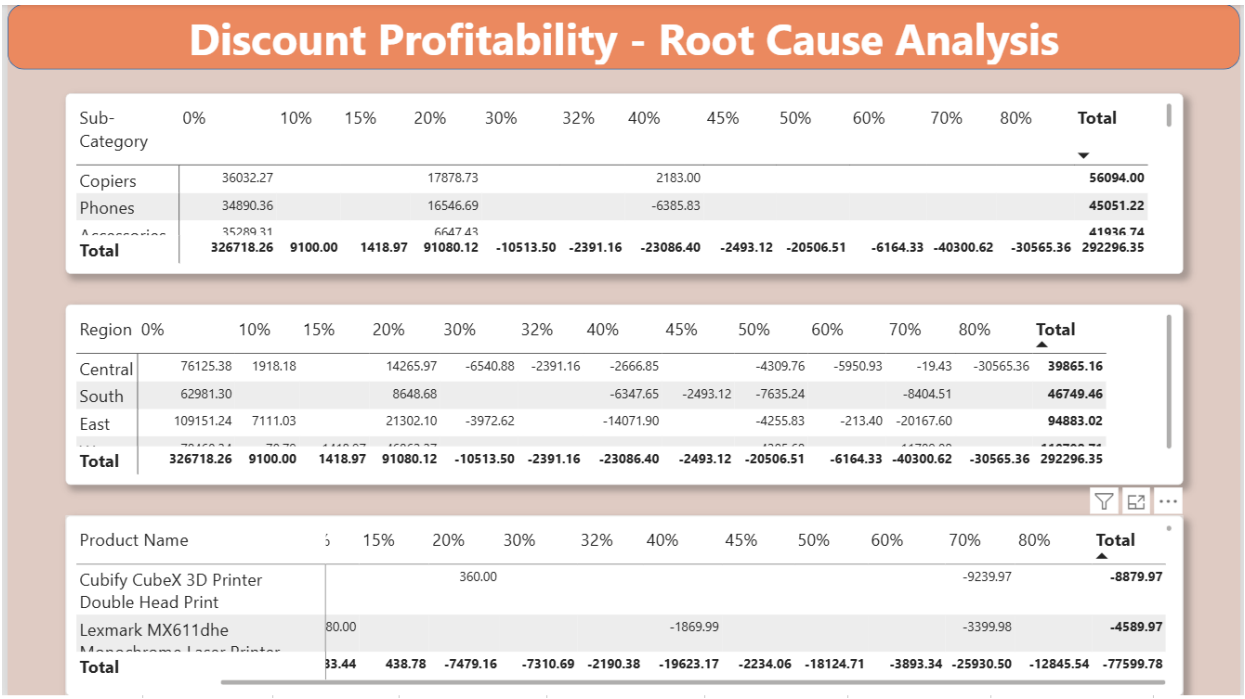
The report uses KPI cards, a sub-category profit bar chart, regional profit chart, discount-level profit and margin visuals, matrix drill-downs, and slicers. The executive overview supports high-level review; discount analysis and root-cause pages enable more detailed investigation.



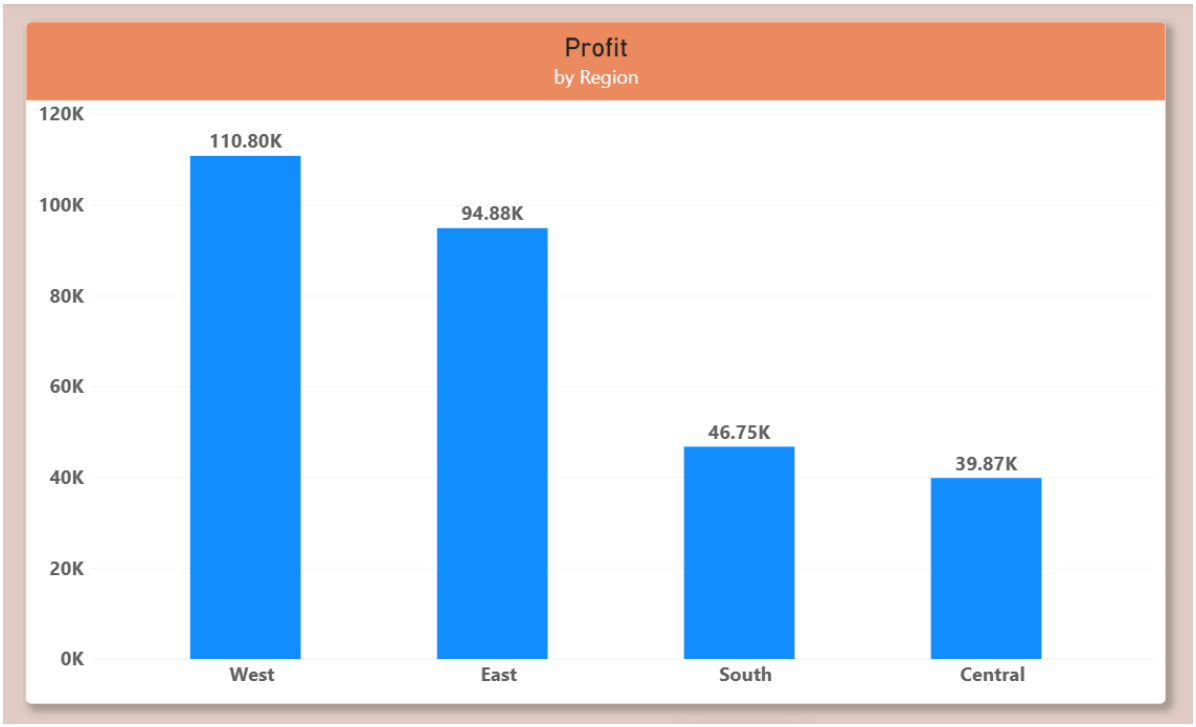
Executive overview



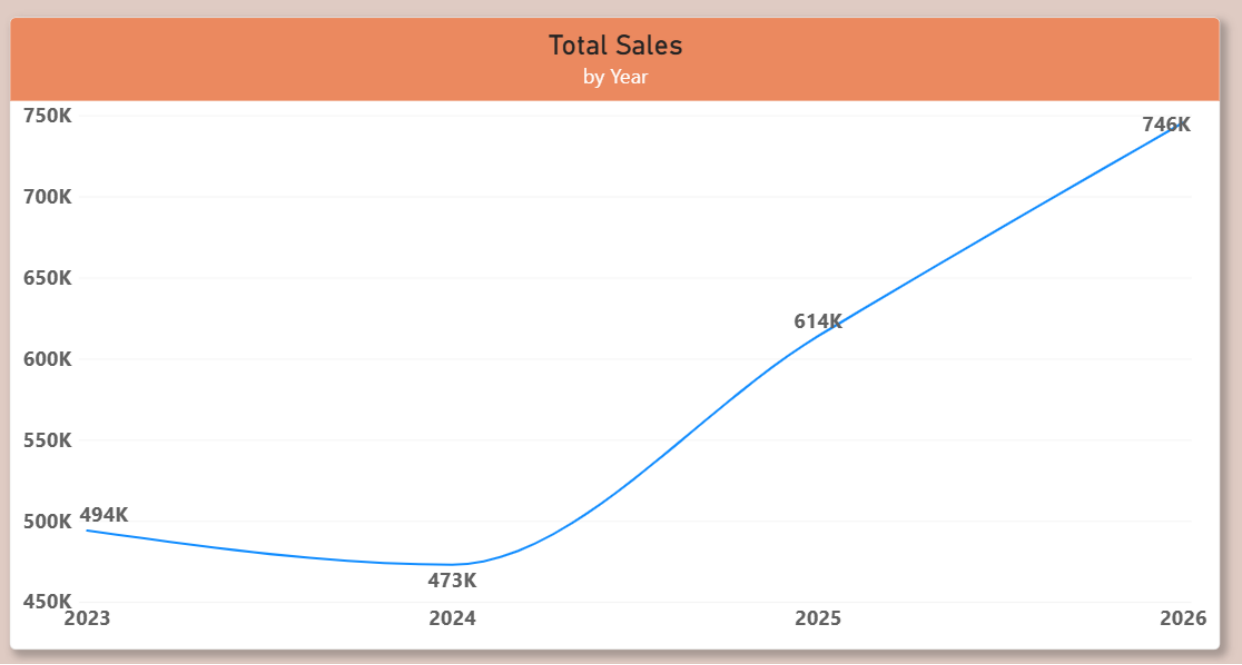
Discount-level profitability



Discount root-cause analysis



Regional profitability



Yearly sales performance

8. Business Recommendations

1. Investigate Tables, Bookcases, and Supplies for pricing, discount, cost, and product-mix factors.
2. Prioritize Furniture margin, especially Tables and Bookcases.
3. Examine high-discount transactions and the Central Furnishings 60% discount cluster.
4. Compare product mix and discount distribution across regions before attributing differences to one factor.
5. Track margin alongside sales growth; 2026 sales and profit increased, but margin was lower than in 2025.
6. Document the exact dataset source, metric definitions, and refresh process.

9. Limitations

• Exact source filename/version and download date were not recorded.
• The analysis notes do not document a separate shipping-cost field.
• Discount and regional patterns are descriptive and not causal proof.
• Returns has one row per returned Order ID and should not be naively merged into order-line data without checking cardinality.
• Some matrix screenshots are clipped; detailed figures rely on recorded analysis notes.
• The recorded date range does not prove full calendar-year completeness.

10. Skills Demonstrated

Data profiling; data-quality checks; Excel formulas and PivotTables; KPI definition; exploratory data analysis; profitability analysis; Power BI data modelling and visualisation; slicer/filter interaction; business interpretation; communicating limitations and recommendations.

11. Conclusion

The project demonstrates an end-to-end descriptive analytics workflow: profiling transactional data, defining KPIs, exploring profitability across multiple dimensions, and presenting results in Power BI. The strongest follow-up work is to investigate Furniture, loss-making sub-categories, and high-discount transactions while validating business context before recommending policy changes.